Viridis Research — certification record

Run `runs/2026-11-01_P2` 2026-11-01 project P2, task P2-T004 engine `9c1495480475`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/41315221075`

Re-certifying the Viridis OS Decision Kernels: A Lean Proof of the Afforestation Seeding Core
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Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**PROVEN**	formal [P2.Afforestation]: (1) For all real a, sigma, Delta mu, t > 0 and every real C, eff(a, sigma, t Delta mu, C) = t^3 eff(a, sigma, Delta mu, C), where eff(a, sigma, Delta mu, C) = C (3 Delta mu / (2 a sigma))^3. (2) In every real inner-product space, for nonzero n and m, eta(n, m) = <n,m>^2 / (|n|^2 |m|^2) satisfies 0 <= eta(n, m) <= 1.
C2	**ASSUMED**	conjecture: Observation from a local check, not a sealed claim: both parts of P2.Afforestation hold without any of their hypotheses. The cubic law is a ring identity for all real a, sigma, Delta mu, t, C, and the alignment bound also holds for n = 0 or m = 0, where eta = 0 by Lean's x/0 = 0 convention.

Sealed statement P2.Afforestation

File `lean/ViridisStatements/P2/Afforestation.lean`, sha256 `2fc1cbb05d7780f266f9da9c00941ca1b7d8ebc703a11a25fc9bb23237a59436`, sealed by decision D-007. Axioms used by the proof: Classical.choice, Quot.sound, propext. Candidate sha256 `a179401aa4d0e0de8a14944fe4356287cdd2295f2d71d3fcd87a8093091ca48c`. Listing shown with symbol substitution; the exact bytes are in the repository.

```
import Mathlib

/-!
# P2 · re-certification: the Afforestation stewardship laws

The live Viridis OS Afforestation kernel ranks seeding plans and site-preparation levers by
seeding efficiency (established canopy per seed) and scores how well a seed allocation is
aligned with the optimal one. Its formal backing is the Afforestation Stewardship Theorem
clean core,
‘Viridis.Afforestation.AfforestationStewardship.establishment_efficiency_cubic_in_drive_over_tension‘
and ‘Viridis.Afforestation.AforestationStewardship.seeding_efficiency_eq_cos2_theta‘, in the
archived canon (github.com/jdhart81/viridis-canon, commit
80d32904d751b45f415e23f872510cc2eddca3ff, file ‘AfforestationStewardship.lean‘, sha256
3ebe33bfd6bd74124e87b5d7e73448db58e1459207fd6ff5c00a90fc3ac9526f).
This module restates those two theorems, over the same definitions, as one sealed target.

Model: for a site with ecological drive ‘Δμ‘, edge hostility ("surface tension") ‘σ‘, shape
constant ‘a‘ and canopy per established nucleus ‘C‘, the seeding efficiency is
‘eff = C·(3Δμ/(2aσ))^3 = C/n*‘, where ‘n* = (2aσ/(3Δμ))^3‘ is the critical nucleus.
The alignment of an allocation ‘n‘ with the optimal seeding vector ‘m‘ is
‘? = ⟨n, m⟩^2 / (‖n‖^2·‖m‖^2)‘ (= cos^2 ?); the forcing debt is ‘1 − ?‘.

‘Statement‘, part 1: for ‘a, σ, Δμ, t > 0‘, scaling the drive by ‘t‘ multiplies the efficiency
by ‘t^3‘. Part 2: in every real inner-product space, for nonzero ‘n, m‘, ‘0 ≤ ? ≤ 1‘.
-/

namespace ViridisStatements.P2.Afforestation

/-- Critical nucleus size ‘n* = (2aσ/(3Δμ))^3‘ (as in the Restoration kernel). -/
noncomputable def nStar (a σ Δμ : ℝ) : ℝ := (2 * a * σ / (3 * Δμ)) ^ 3

/-- Seeding efficiency ‘e = C / n* = C·(3Δμ/(2aσ))^3‘ (established canopy per seed). -/
noncomputable def eff (a σ Δμ C : ℝ) : ℝ := C * (3 * Δμ / (2 * a * σ)) ^ 3

/-- Alignment of a seed allocation ‘n‘ with the optimal seeding vector ‘m‘:
‘? = ⟨n, m⟩^2 / (‖n‖^2·‖m‖^2)‘, the squared cosine of the angle between them. -/
noncomputable def eta {E : Type*} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
    (n m : E) : ℝ :=
  (inner ℝ n m : ℝ) ^ 2 / (‖n‖ ^ 2 * ‖m‖ ^ 2)

/-- **Sealed target ‘P2.Afforestation‘.**
(1) Cubic establishment-efficiency law: for ‘a, σ, Δμ, t > 0‘ and any ‘C‘,
‘eff a σ (t·Δμ) C = t^3 · eff a σ Δμ C‘.
(2) Seeding alignment is a squared cosine: for nonzero ‘n, m‘ in any real inner-product
```

```
space, `0 ≤ ?(n, m) ≤ 1`. -/
def Statement : Prop :=
  (∀ a σ Δμ C t : ℝ, 0 < a → 0 < σ → 0 < Δμ → 0 < t →
    eff a σ (t * Δμ) C = t ^ 3 * eff a σ Δμ C) ∧
  (∀ (E : Type*) [NormedAddCommGroup E] [InnerProductSpace ℝ E] (n m : E),
    n ≠ 0 → m ≠ 0 → 0 ≤ eta n m ∧ eta n m ≤ 1)

/-! ## Sanity checks

These pin the meaning of the definitions with concrete numbers. -/

/-- Efficiency is canopy per seed: with `a = σ = Δμ = 1`, `C = 1`, `n* = 8/27` and
`eff = 27/8 = C / n*`. -/
example : nStar 1 1 1 = 8/27 ∧ eff 1 1 1 1 = 27/8 ∧ eff 1 1 1 1 = 1 / nStar 1 1 1 := by
  norm_num [nStar, eff]

/-- Doubling the drive multiplies efficiency by `2^3 = 8`: `eff(Δμ = 2) = 27 = 8·(27/8)`. -/
example : eff 1 1 2 1 = 27 ∧ eff 1 1 2 1 = 2 ^ 3 * eff 1 1 1 1 := by
  norm_num [eff]

/-- Negative control: the law is cubic, not linear: doubling the drive does not merely
double the efficiency. -/
example : eff 1 1 2 1 ≠ 2 * eff 1 1 1 1 := by
  norm_num [eff]

/-- Halving the edge hostility `σ` gives the canon's 8× site-prep lever at a concrete point. -/
example : eff 1 (1/2) 1 1 = 8 * eff 1 1 1 1 := by
  norm_num [eff]

/-- Parallel allocations are perfectly aligned: in `ℝ`, `?(2, 3) = 1`. -/
example : eta (2 : ℝ) 3 = 1 := by
  norm_num [eta]

/-- Orthogonal allocations have zero alignment: in `ℝ²`, `?((1,0), (0,1)) = 0`. -/
example : eta (!₂[1, 0] : EuclideanSpace ℝ (Fin 2)) !₂[0, 1] = 0 := by
  simp [eta, EuclideanSpace.inner_eq_star_dotProduct, dotProduct, Fin.sum_univ_two]

/-- Negative control: without the normalisation the squared inner product is not bounded by 1:
in `ℝ`, `⟨2, 3⟩² = 36 > 1`, while `?(2, 3) = 1`. -/
example : (1 : ℝ) < (inner ℝ (2 : ℝ) 3) ^ 2 := by
  norm_num

end ViridisStatements.P2.Afforestation
```

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 5 entries, 5 DOIs, 5 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 0.

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED/REPLICATED**: reproduced by the verifier (not yet enabled). **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

Re-certifying the Viridis OS Decision Kernels: Carbon Continuity, the Restoration Critical Nucleus, and the Afforestation Seeding Core

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Abstract

Viridis OS runs decision kernels whose formal cores were checked by tools that can no longer be re-run. This paper documents the project that brings each kernel to a re-checkable standard: the kernel's formal core is restated as a sealed, self-contained Lean 4 statement and re-proved under a pinned toolchain. The first kernel, *Carbon Continuity*, has a Lean proof of its sealed statement: a two-pool carbon portfolio can be kept from shrinking exactly when $(1-a)(1-d) \leq rp$. The second kernel, *Restoration*, decides whether a planted cluster on degraded land is large enough to grow into forest; we give a Lean proof of the sealed statement `P2.Restoration`. The critical radius $r^\star = 2a\sigma/(3\Delta\mu)$ is positive and is the unique maximiser of the free energy $G(r) = -\Delta\mu r^3 + a\sigma r^2$ on $r \geq 0$, and the barrier satisfies $G(r^\star) = \frac{1}{2}\Delta\mu(r^\star)^3$. The proof rests on the factorisation $G(r^\star) - G(r) = \Delta\mu(r - r^\star)^2(r + r^\star/2)$. We set out what this settles (the go/no-go geometry inside the nucleation model) and what it leaves to field evidence (parameter values, adequacy of the model, rates, seed allocation and code conformance). This revision re-certifies the third kernel, *Afforestation*, which ranks seeding plans by seeding efficiency and scores how well a seed allocation is aligned with the optimal one. We give a Lean proof of the sealed statement `P2.Afforestation`: seeding efficiency $\mathrm{eff} = C\,(3\Delta\mu/(2a\sigma))^3$ scales as the cube of the ecological drive $\Delta\mu$, and the seeding alignment $\eta = \langle n, m\rangle^2/(\|n\|^2\|m\|^2)$ lies in $[0, 1]$ in every real inner-product space. The first part is a ring identity and the second is the squared Cauchy–Schwarz inequality. We observe that none of the hypotheses of either part is needed, that the canon proof still replays verbatim, and that the canon's "$\cos^2\Theta$" name promises more than the bound that is proved. The statement proposals of 30 October are retained. `P2.Harmonization` restates two properties of a scalar-price welfare theorem for N quadratic stewards: market clearing, and the absence of a duality gap. Each proposal copies the canon definitions and hypotheses verbatim and records the hash of the canon file. It also includes sanity examples with concrete numbers, among them negative controls. We point out where canon theorem names promise more than the theorems prove. We also point out which canon hypotheses are kept for faithfulness although the proofs do not need them. `P2.Restoration` and `P2.Afforestation` have since been sealed unchanged and are proved here. No proof of `P2.Harmonization` is claimed.

1 Introduction

Viridis OS runs a set of decision kernels. Each one is a small piece of software that turns ecological parameters into a decision. A kernel's rule is only as trustworthy as the mathematics behind it. The archived canon [3] proved those rules in Lean, but the certificates came from tooling that has since been retired. The project described here moves each kernel's assurance onto a re-checkable

footing in two steps. First, a *sealed statement* fixes the exact proposition a kernel relies on. Second, a Lean proof of that statement is checked under Lean v4.28.0 with a pinned Mathlib [5].

The two steps are separate on purpose. Writing the statement is where most of the risk lies. A statement can be correct and easy to prove and still be the wrong one. It can be weaker than what the kernel uses, stronger than what the canon proved, or vacuous. The 30 October revision covered that first step for three kernels: Restoration, Afforestation and Harmonization. The 31 October revision took the second step for Restoration (Section 4), and the present revision takes it for Afforestation (Section 5). The Restoration kernel makes a go/no-go call for a planted cluster. Go means the cluster is past the nucleation barrier and will grow into forest. No-go means it is still on the uphill side and is expected to dissolve. A wrong rule here would send planting budgets to clusters that cannot persist, so the geometry of the barrier is worth checking mechanically, however elementary it is. The Afforestation kernel works one step later: given that clusters must clear the barrier, it compares seeding plans and site-preparation levers (such as nurse plants or mulch, which lower the edge hostility σ) by how much established canopy each seed buys, and it scores a proposed seed allocation by its alignment with the optimal one. Both rules rest on two short facts, a scaling law and a Cauchy–Schwarz bound, which are what `P2.Afforestation` seals. The question is narrow. We ask what each kernel's rule *mathematically* relies on, and how to write that down self-contained and faithful to the canon theorem it replaces. We do not ask whether the models describe any real landscape.

2 Background and related work

Carbon Continuity. One disturbance-recovery cycle maps $(L, D) \mapsto (aL + rD,\ pL + dD)$. This is the nonnegative matrix $M = \begin{pmatrix} a & r \\ p & d \end{pmatrix}$, and the threshold theorem is a two-dimensional instance of the Collatz–Wielandt characterisation in Perron–Frobenius theory [1].

Restoration and Afforestation. The canon models a planted cluster as a nucleus of the forest phase inside a metastable degraded phase. It uses the free-energy form of classical nucleation theory [4]: a bulk gain $-\Delta\mu\, n$ competes with a surface cost $a\sigma\, n^{2/3}$. In the cube-root variable $r = n^{1/3}$ this becomes the cubic $G(r) = -\Delta\mu r^3 + a\sigma r^2$. The textbook facts are a critical size at the barrier top and a barrier equal to half the bulk term at the critical size. These are the mathematical content of `P2.Restoration`. The Afforestation efficiency $e = C/n^\star$ inherits the cube from $n^\star = (r^\star)^3$. The alignment score is a squared cosine, bounded by Cauchy–Schwarz.

Harmonization. The scalar-price theorem is the separable quadratic case of Lagrangian duality with one linear coupling constraint [2]. At the clearing price, the decentralised best responses solve the centralised problem, and there is no duality gap.

We claim no novelty for any of this mathematics. The contribution is a set of faithful, machine-checkable targets tied to deployed kernels.

3 Carbon Continuity (P2-T001)

The sealed module defines $\mathrm{stepLiving}(a, r, L, D) = aL + rD$ and $\mathrm{stepDurable}(d, p, L, D) = pL + dD$. $\mathrm{Sustainable}(a, d, r, p)$ means that some $L, D > 0$ satisfy $L \le aL + rD$ and $D \le pL + dD$.

Theorem 1 (`P2.CarbonContinuity`). *For all real a, d, r, p with $0 \le a < 1$, $0 \le d < 1$, $r > 0$ and $p > 0$:* $\mathrm{Sustainable}(a, d, r, p) \iff (1 - a)(1 - d) \le rp$.

Proof sketch. Sufficiency uses the witness $(L, D) = (r, 1-a)$. The living step is then an identity, and the durable step reduces to $rp-(1-a)(1-d) \geq 0$. For necessity, rearrange the two inequalities to $(1-a)L \leq rD$ and $(1-d)D \leq pL$, multiply them (both sides are positive), and cancel $LD > 0$. □

The previous revision gave a Lean proof of this statement (file `proofs/P2.CarbonContinuity.lean` of run 2026-10-29), with axioms `propext`, `Classical.choice` and `Quot.sound`.

```
theorem main : ViridisStatements.P2.CarbonContinuity.Statement := by
  intro a d r p _ ha1 _ hd1 hr _
  exact ⟨necessary a d r p ha1 hd1, sufficient a d r p ha1 hr⟩
```

We observe that the helper lemmas use only $a < 1$, $d < 1$ and $r > 0$.

4 Restoration: the critical nucleus (P2-T003)

4.1 The sealed statement

The sealed module `ViridisStatements.P2.Restoration` (sha256 `fd40b113...ccff18`) is the statement proposed on 30 October (Section 7), unchanged. It defines $r^\star(\Delta\mu, a, \sigma) = 2a\sigma/(3\Delta\mu)$ and $G(\Delta\mu, a, \sigma, r) = -\Delta\mu r^3 + a\sigma r^2$ verbatim from the canon file `ForestNucleation.lean` (sha256 `667a826a...ec28132`, which we checked against the pinned manifest).

Theorem 2 (`P2.Restoration`). *(i) For all real $\Delta\mu, a, \sigma > 0$: $r^\star > 0$, $G(r) \leq G(r^\star)$ for every $r \geq 0$, and every $r \geq 0$ with $G(r) = G(r^\star)$ equals $r^\star$.*

(ii) For all real $\Delta\mu > 0$ and all real a, σ: $G(r^\star) = \frac{1}{2}\Delta\mu\,(r^\star)^3$.

We give a Lean proof of Theorem 2 in `proofs/P2.Restoration.lean`. The local checker, which runs the same stages as the verifier, passes it, and the only axioms it reports are `propext`, `Classical.choice` and `Quot.sound`.

4.2 Proof

Proof. Let $\Delta\mu \neq 0$. By the definition of $r^\star$, $a\sigma = \frac{3}{2}\Delta\mu\, r^\star$. Substituting into G gives $G(r) = \Delta\mu\big(-r^3 + \frac{3}{2}r^\star r^2\big)$, hence $G(r^\star) = \frac{1}{2}\Delta\mu(r^\star)^3$, which is (ii). Subtracting,

$$G(r^\star) - G(r) = \Delta\mu\big(r^3 - \tfrac{3}{2}r^\star r^2 + \tfrac{1}{2}(r^\star)^3\big) = \Delta\mu\,(r - r^\star)^2\,(r + r^\star/2).$$

Now let $\Delta\mu, a, \sigma > 0$. Then $r^\star > 0$ as a quotient of positive numbers. For $r \geq 0$ all three factors on the right are nonnegative, so $G(r) \leq G(r^\star)$. If equality holds, the product is zero. Since $\Delta\mu > 0$ and $r + r^\star/2 > 0$, we get $(r - r^\star)^2 = 0$, that is, $r = r^\star$. □

The Lean development follows this argument line by line. Its key lemmas are:

```
lemma aσ_eq (Δμ a σ: ℝ) (hΔμ : Δμ≠0) :
    a * σ= 3 / 2 * Δμ* rStar Δμa σ:= by
  unfold rStar
  field_simp

lemma key_identity (Δμ a σr : ℝ) (hΔμ : Δμ≠0) :
    G Δμa σ(rStar Δμa σ) - G Δμa σr =
      Δμ* (r - rStar Δμa σ) ^ 2 * (r + rStar Δμa σ/ 2) := by
  unfold G
```

```
  rw [aσ_eq Δμa σhΔμ]
  ring

lemma barrier_half (Δμ a σ: ℝ) (hΔμ : 0 < Δμ) :
    G Δμa σ(rStar Δμa σ) = (1 / 2) * Δμ* (rStar Δμa σ) ^ 3 := by
  unfold G
  rw [aσ_eq Δμa σhΔμ.ne']
  ring

theorem main : ViridisStatements.P2.Restoration.Statement :=
  ⟨fun Δμa σhΔμ ha hσ => critical_nucleus Δμa σhΔμ ha hσ,
   fun Δμa σhΔμ => barrier_half Δμa σhΔμ⟩
```

Here `critical_nucleus` proves (i). Maximality comes from `key_identity`, `positivity` and `linarith`. Uniqueness is a two-way `mul_eq_zero` case split followed by `pow_eq_zero_iff`.

4.3 Relation to the canon proof

The canon proof uses the same factorisation, but in a compressed tactic style (`refine'`, a long term-mode `contrapose!` chain, and `grind` for the half identity). We observe that the canon proofs, pasted verbatim into a candidate file, also pass the local checker under the pinned toolchain. The archived proof therefore still replays. We submit the restructured version because its helper lemmas name the single algebraic fact everything rests on, and because it depends less on the behaviour of `grind`, which may change between Mathlib versions.

4.4 Observations

- **An unused hypothesis.** Part (ii) holds for every real $\Delta\mu$. For $\Delta\mu \neq 0$ the argument above does not use the sign. For $\Delta\mu = 0$, Lean's convention $x/0 = 0$ gives $r^\star = 0$, and both sides are 0. We checked both cases locally. The sealed statement keeps $0 < \Delta\mu$ to match the canon.
- **The sign of $\Delta\mu$ matters for (i).** If $\Delta\mu < 0$ the degraded state is favoured, $r^\star < 0$, and G is unbounded above on $r \geq 0$, so no cluster size is a barrier top. The sealed module includes a negative control for this case.
- **r versus n.** The theorem concerns the cube-root variable $r = n^{1/3}$. Since $r \mapsto r^3$ is an increasing bijection of $[0, \infty)$, it follows informally that $n^\star = (r^\star)^3$ is the unique maximiser of $-\Delta\mu n + a\sigma n^{2/3}$ on $n \geq 0$. This translation is not part of the sealed statement, and it has no Lean proof here.

5 Afforestation: the seeding core (P2-T004)

5.1 The sealed statement

The sealed module `ViridisStatements.P2.Afforestation` (sha256 `2fc1cbb0...59436`) is byte-identical to the statement proposed on 30 October (Section 7). It defines $n^\star(a, \sigma, \Delta\mu) = (2a\sigma/(3\Delta\mu))^3$, $\mathrm{eff}(a, \sigma, \Delta\mu, C) = C\,(3\Delta\mu/(2a\sigma))^3$ and, on any real inner-product space E, $\eta(n, m) = \langle n, m\rangle^2/(\|n\|^2\|m\|^2)$. The first two are verbatim from the canon file `AforestationStewardship.lean` (sha256 `3ebe33bf...9526f`, checked against the pinned manifest); η names the expression in the canon theorem's conclusion.

Theorem 3 (`P2.Afforestation`). *(i) For all real $a, \sigma, \Delta\mu, t > 0$ and every real C:* $\operatorname{eff}(a, \sigma, t\Delta\mu, C) = t^3 \operatorname{eff}(a, \sigma, \Delta\mu, C)$.

(ii) For every real inner-product space E and all $n, m \in E$ with $n \neq 0$, $m \neq 0$: $0 \leq \eta(n, m) \leq 1$.

We give a Lean proof of Theorem 3 in `proofs/P2.Afforestation.lean`. The local checker passes it, and the only axioms it reports are `propext`, `Classical.choice` and `Quot.sound`.

5.2 Proof

Proof. (i) $C\,(3t\Delta\mu/(2a\sigma))^3 = t^3 \cdot C\,(3\Delta\mu/(2a\sigma))^3$ is an identity of rational expressions in which t appears only in the numerator. (ii) η is a square divided by a square, so $\eta \geq 0$. By Cauchy–Schwarz, $|\langle n, m\rangle| \leq \|n\|\,\|m\|$; both sides are nonnegative, so squaring gives $\langle n, m\rangle^2 \leq \|n\|^2\|m\|^2$, and dividing by the nonnegative denominator gives $\eta \leq 1$. □

In Lean, division by zero is defined to be zero, so both steps go through without the hypotheses: for (i), `ring` treats x/y as $x \cdot y^{-1}$; for (ii), `div_le_one_of_le₀` needs only a nonnegative denominator. The submitted file therefore proves hypothesis-free helper lemmas and discards the hypotheses in `main`:

```
theorem eff_cubic (a σΔμC t : ℝ) :
    eff a σ(t * Δμ) C = t ^ 3 * eff a σΔμC := by
  unfold eff
  ring

theorem eta_mem_unit {E : Type*} [NormedAddCommGroup E] [InnerProductSpace ℝE]
    (n m : E) : 0 ≤eta n m ∧eta n m ≤1 := by
  unfold eta
  refine ⟨by positivity, ?_⟩
  apply div_le_one_of_le₀ _ (by positivity)
  have h := abs_real_inner_le_norm n m
  have h2 : |inner ℝn m| ^ 2 ≤(‖n‖ * ‖m‖) ^ 2 :=
    pow_le_pow_left₀ (abs_nonneg _) h 2
  rw [sq_abs, mul_pow] at h2
  exact h2

theorem main : ViridisStatements.P2.Afforestation.Statement := by
  refine ⟨fun a σΔμC t _ _ _ _ => eff_cubic a σΔμC t, ?_⟩
  intro E _ _ n m _ _
  exact eta_mem_unit n m
```

5.3 Relation to the canon proof

Part (i) is the canon proof unchanged (`unfold eff; ring`). For part (ii) the canon closes the numerator bound with `nlinarith` fed the two sides of $|\langle n, m\rangle| \leq \|n\|\|m\|$; we replace that search by an explicit squaring step (`pow_le_pow_left₀`, `sq_abs`, `mul_pow`), which does not depend on `nlinarith`'s heuristics. We observe that the canon proofs, pasted verbatim (including `refine'`) into a candidate file, also pass the local checker under the pinned toolchain. The sealed statement quantifies over a universe-polymorphic type `E : Type*`, as the canon does; the polymorphic candidate passed the statement-match stage.

5.4 Observations

- **No hypothesis is used.** The cubic law holds for all real $a, \sigma, \Delta\mu, t, C$ (as a Lean identity; when $a\sigma = 0$ both sides are 0). The alignment bound holds for $n = 0$ or $m = 0$, where $\eta = 0/0 = 0$ in Lean. The sealed statement keeps all hypotheses to match the canon. The helper lemmas in the submitted file state the hypothesis-free versions; they are not sealed statements.

- **The name over-promises.** The canon theorem is called `seeding_efficiency_eq_cos2_theta`, and its docstring describes $\eta = \cos^2\Theta$ with forcing debt $1 - \eta = \sin^2\Theta$ and $\eta = 1$ exactly for aligned allocations. Only the bound $0 \le \eta \le 1$ is proved, in the canon and here. A kernel that reports "zero forcing debt iff the allocation is optimal" relies on the equality case of Cauchy–Schwarz, which is not sealed.

- **Scale invariance, not proved.** $\eta(cn, m) = \eta(n, m)$ for $c \neq 0$, so the alignment score ignores the total seed budget and measures only the direction of the allocation. This is immediate but has no Lean proof here.

- **The site-prep lever.** The canon's $8\times$ lever (halving σ multiplies eff by 8) is the same kind of ring identity; it is pinned in the sealed module only by a concrete `example`, not as a general statement.

6 Method for writing a sealed statement

For each kernel we followed the same procedure.

1. **Identify the canon theorems.** We take the theorems named in the task brief and check the canon file's SHA-256 against the pinned manifest (canon commit `80d32904`).

2. **Copy definitions verbatim.** We copy only the definitions the theorems use, with the same argument order, and drop the rest. Any added abbreviation must unfold to the canon expression exactly.

3. **Copy hypotheses verbatim.** Each conjunct of `Statement` keeps exactly the hypotheses of its canon theorem. We drop no hypothesis, even one that is mathematically unnecessary, because dropping it would strengthen the canon statement. We add none, because adding one would weaken it.

4. **State the conclusion as proved, not as named.** Where a canon docstring or name claims more than the Lean conclusion, the statement follows the conclusion.

5. **Pin the meaning.** We add sanity `example`s with concrete numbers, closed by `norm_num` or `simp` (`decide` does not apply to $\mathbb{R}$). They include negative controls showing that the key hypotheses or factors cannot be dropped.

6. **Test non-triviality.** We compile each module and probe `Statement` with 28 one-shot tactics after `unfold Statement`. The tactics include `rfl`, `decide`, `simp`, `norm_num`, `aesop`, `linarith`, `nlinarith`, `positivity`, `grind`, `field_simp`, `ring` and `polyrith`.

7. **Test feasibility.** Privately, we check that a port of the canon proof closes the proposed statement. This guards against proposing a false target. It is not a claim of proof.

7 The three proposals (P2-T002)

The three statements below were proposed on 30 October. `P2.Restoration` has since been sealed unchanged and is proved in Section 4; `P2.Afforestation` has been sealed unchanged and is proved in Section 5. `P2.Harmonization` awaits its proof task.

7.1 `P2.Restoration`: the critical-nucleus go/no-go

Definitions: $r^\star(\Delta\mu, a, \sigma) = 2a\sigma/(3\Delta\mu)$ and $G(\Delta\mu, a, \sigma, r) = -\Delta\mu r^3 + a\sigma r^2$.

Proposed statement 4. (i) For all $\Delta\mu, a, \sigma > 0$: $r^\star > 0$; $G(r) \le G(r^\star)$ for all $r \ge 0$; and $G(r) = G(r^\star)$ with $r \ge 0$ implies $r = r^\star$. (ii) For all $\Delta\mu > 0$ and all a, σ: $G(r^\star) = \frac{1}{2}\Delta\mu\,(r^\star)^3$.

Part (i) restates canon `critical_nucleus`, and part (ii) restates `barrier_half_identity` (file sha256 `667a826a...ec28132`). Both follow from the factorisation $G(r^\star) - G(r) = \Delta\mu\,(r - r^\star)^2\,(r + r^\star/2)$. The kernel's go/no-go rule compares cluster size with $n^\star = (r^\star)^3$. Below $n^\star$ the cluster is on the uphill side of the barrier. Negative controls: $G(-1) = 5/2 > G(r^\star) = 1/2$ at $(\Delta\mu, a, \sigma) = (1, 1, 3/2)$, so the restriction $r \ge 0$ is essential. $G(r^\star) \ne \Delta\mu(r^\star)^3$, so the factor $\frac{1}{2}$ is substantive. $\Delta\mu = -1$ gives $r^\star < 0$.

```
def Statement : Prop :=
  (∀ Δμa σ: ℝ, 0 < Δμ→0 < a →0 < σ→
    0 < rStar Δμa σ∧
    (∀ r : ℝ, 0 ≤r →G Δμa σr ≤G Δμa σ(rStar Δμa σ)) ∧
    (∀ r : ℝ, 0 ≤r →G Δμa σr = G Δμa σ(rStar Δμa σ) →r = rStar Δμa σ)) ∧
  (∀ Δμa σ: ℝ, 0 < Δμ→
    G Δμa σ(rStar Δμa σ) = (1 / 2) * Δμ* (rStar Δμa σ) ^ 3)
```

7.2 `P2.Afforestation`: cubic efficiency and seeding alignment

Definitions: $\mathrm{eff}(a, \sigma, \Delta\mu, C) = C\,(3\Delta\mu/(2a\sigma))^3$ and $\eta(n, m) = \langle n, m\rangle^2/(\|n\|^2\|m\|^2)$.

Proposed statement 5. (i) For all $a, \sigma, \Delta\mu, t > 0$ and all C: $\mathrm{eff}(a, \sigma, t\Delta\mu, C) = t^3\,\mathrm{eff}(a, \sigma, \Delta\mu, C)$. (ii) In every real inner-product space E (universe-polymorphic, as in the canon), for $n, m \ne 0$: $0 \le \eta(n, m) \le 1$.

These restate the canon theorems `establishment_efficiency_cubic_in_drive_over_tension` and `seeding_efficiency_eq_cos2_theta` (file sha256 `3ebe33bf...9526f`).

Remark 6. The second canon theorem is named "eff $= \cos^2\Theta$". Its docstring also mentions $\eta = 1$ iff $n \parallel m$, and $1 - \eta = \sin^2\Theta$. Its Lean conclusion proves only the bound $0 \le \eta \le 1$, and the proposal restates only that bound. If the kernel uses the equality case, that case needs its own statement.

Sanity examples pin eff $= C/n^\star$ (27/8 at unit parameters), the factor 8 when the drive is doubled, and the factor 8 when σ is halved. They also pin $\eta = 1$ for parallel and $\eta = 0$ for orthogonal vectors in $\mathbb{R}^2$. Negative controls: doubling the drive does not double the efficiency, and the unnormalised $\langle 2, 3\rangle^2 = 36$ exceeds 1.

```
def Statement : Prop :=
  (∀ a σΔμC t : ℝ, 0 < a →0 < σ→0 < Δμ→0 < t →
    eff a σ(t * Δμ) C = t ^ 3 * eff a σΔμC) ∧
  (∀ (E : Type*) [NormedAddCommGroup E] [InnerProductSpace ℝE] (n m : E),
    n ≠0 →m ≠0 →0 ≤eta n m ∧eta n m ≤1)
```

7.3 `P2.Harmonization`: one price suffices

Definitions: $\mathrm{br}(\lambda)_i = b_i - \lambda/a_i$, $J(x) = \sum_i \frac{a_i}{2}(x_i - b_i)^2$, $\lambda^\star = (\sum_i b_i - X)/\sum_i a_i^{-1}$ and $x_i^\star = \mathrm{br}(\lambda^\star)_i$.

Proposed statement 7. For all $N > 0$, all $a, b \in \mathbb{R}^N$ with every $a_i > 0$, and all X: (i) $\sum_i \mathrm{br}(\lambda^\star)_i = X$; and (ii) every y with $\sum_i y_i = X$ satisfies $J(x^\star) \le J(y)$.

These restate `ght_market_clears` and `ght_strong_duality_decentralized_eq_centralized` (file sha256 `cfdead65...110aec`). On the canon instance $N = 2$, $a = b = (1, 1)$, $X = 0$, the sanity examples give $\lambda^\star = 1$, $x^\star = (0, 0) \neq b$, $J(x^\star) = 1$, and $J(1, -1) = 2$. Negative controls:

- The infeasible ideal $y = b$ has $J = 0 < 1$, so the feasibility hypothesis is essential.
- The price $\lambda = 0$ does not clear.
- With $N = 0$ the aggregate is $0 \neq X = 1$, so $N > 0$ is essential.

```
def Statement : Prop :=
  ∀(N : ℕ) (a b : Fin N →ℝ), 0 < N →(∀ i, 0 < a i) →∀X : ℝ,
    (∑ i, br a b (lamStar a b X) i) = X ∧
    (∀ y : Fin N →ℝ, (∑ i, y i) = X →J a b (xStar a b X) ≤J a b y)
```

8 Results

Kernel	Sealed statement	Session	Status of our candidate
Carbon Continuity	`P2.CarbonContinuity`	2026-10-29	Lean proof, local check passed
Restoration	`P2.Restoration`	2026-10-31	Lean proof, local check passed
Afforestation	`P2.Afforestation`	2026-11-01	Lean proof, local check passed
Harmonization	proposed only	2026-10-30	no proof yet

Table 1: State of the P2 program. "Local check passed" means `viridis-lean-check` accepted the file, with axioms `propext`, `Classical.choice` and `Quot.sound`. The verifier awards the labels.

For Afforestation, the results of 1 November are as follows. The restructured proof of `P2.Afforestation` passed the local checker on the first attempt. The verbatim canon proof also passed it. Neither part needs any of its hypotheses. Aristotle was not used.

For Restoration, the results of 31 October were as follows. The restructured proof of `P2.Restoration` passed the local checker on the first attempt. The verbatim canon proof also passed it. The half-barrier identity holds without its hypothesis $0 < \Delta\mu$. Aristotle was not used.

The 30 October proposals are summarised below. We observe the following.

- For each proposal, a private port of the canon proof elaborated against a local build of the module. It depends only on `propext`, `Classical.choice` and `Quot.sound`. We do not present these as proofs of the proposals. The proofs belong to later tasks, against the sealed modules.
- Several canon hypotheses are not needed mathematically but are kept for faithfulness:
  - In Lean's convention $x/0 = 0$, the half-barrier identity also holds at $\Delta\mu = 0$.
  - The cubic law is a ring identity, so it needs none of its positivity hypotheses.
  - The alignment bound holds trivially when $n = 0$ or $m = 0$, since then $\eta = 0$.
- One canon name over-promises: the "$\cos^2\Theta$" theorem proves only the bound $0 \le \eta \le 1$.

Proposal	Canon theorems restated	Canon file sha256 (prefix)	Probes closed
`P2.Restoration`	`critical_nucleus`, `barrier_half_identity`	`667a826a`	0/28
`P2.Afforestation`	`establishment_efficiency_...`, `seeding_efficiency_...`	`3ebe33bf`	0/28
`P2.Harmonization`	`ght_market_clears`, `ght_strong_duality_...`	`cfdead65`	0/28

Table 2: Proposed sealed statements. All three modules compile under the pinned toolchain, and their sanity examples elaborate. The probe column counts how many of our 28 one-shot tactics closed `Statement` locally. The verifier runs its own probes.

9 What the statements settle and what they do not

- **Settled, once sealed and proved:** the internal mathematics of each kernel's rule inside its stated model.
- **Not settled:** any parameter value. This covers $\Delta\mu$, σ, a, C, the curvatures a_i, the ideals b_i and the target X. These come from field evidence and remain ASSUMED.
- **Not settled:** whether the models are adequate. The models in question are classical nucleation for tree-cluster establishment and quadratic dissipation for stewardship coordination. Nucleation rates, ecological temperature and stochastic establishment are outside the statements.
- **Not settled:** code conformance. The kernels' runtime identifiers and TypeScript sources were not among this session's inputs.

Restoration in particular. Theorem 2 settles the following:

- Inside the cubic model with positive $\Delta\mu, a, \sigma$ there is exactly one barrier top on $r \geq 0$, and it sits at $r^\star = 2a\sigma/(3\Delta\mu)$.
- G is at most $G(r^\star)$ everywhere on $r \geq 0$.
- The barrier height is $\frac{1}{2}\Delta\mu(r^\star)^3$.

So a kernel that computes $r^\star$ (or $n^\star = (r^\star)^3$) by this formula and calls "go" above it and "no-go" below it is applying a correct consequence of the model. The theorem does *not* settle any of the following, all of which stay ASSUMED:

- the field values of the driving force $\Delta\mu$, the edge "surface tension" σ and the shape factor a, and their uncertainty. Because $n^\star$ scales as $(a\sigma/\Delta\mu)^3$, relative errors in these inputs are roughly tripled in $n^\star$;
- whether classical nucleation theory, with an $n^{2/3}$ edge cost, describes tree-cluster establishment at a given site;
- the time-scale and probability layer: the Arrhenius nucleation rate, the ecological temperature, the stochastic committor that replaces the sharp threshold in practice, and any planting deadline;

- the seed-allocation result `broadcast_suboptimal`, which is in the same canon file but not in the sealed statement;
- the translation from r to cluster size n (a short argument, not formalised here);
- that the deployed code computes $r^\star$ with this formula and applies the stated comparison.

Afforestation in particular. Theorem 3 settles that, inside the model, doubling the ecological drive multiplies seeding efficiency by exactly 8, and more generally by t^3; so a kernel that ranks plans by eff will see drive (and, by the same algebra, edge hostility) changes amplified cubically. It also settles that the alignment score is a number in $[0, 1]$, so the forcing debt $1 - \eta$ is also in $[0, 1]$. It does *not* settle, and the following stay ASSUMED:

- the field values of a, σ, $\Delta\mu$ and the canopy yield C; because of the cube, relative errors in $\Delta\mu/(a\sigma)$ are roughly tripled in eff;
- whether "canopy per seed" $C/n^\star$ is the right ranking criterion, and whether the nucleation model applies to the site;
- the equality case $\eta = 1 \iff n \parallel m$ and the scale invariance of η;
- how the optimal seeding vector m is computed, which is outside the statement;
- the other canon results in the same file (the homogeneous-limit barrier identity and the information-rate ceiling);
- that the deployed code computes eff and η by these formulas.

10 Limitations

The mathematics is elementary, and its value lies in being faithful and re-checkable. Faithfulness was judged by hand against the canon text. No tool checks that a proposal matches its canon theorem. Our probe list is our own guess at the verifier's and may differ from it. `P2.Afforestation` is universe-polymorphic, following the canon's `Type*`. We checked locally that a polymorphic candidate `theorem main : Statement` and the verifier's statement-match template both elaborate. A universe-monomorphic seal (`Type`) would be a mild weakening. The project README asks readers to consult a guidance file (`STATEMENT_PROPOSAL.md`) that was not among the inputs. We followed the brief and the format of the existing sealed module instead. That file was again absent on 31 October and 1 November. For Restoration, the sealed statement quantifies over the cube-root variable r, while field practice counts trees n. The step between the two is informal in this paper.

11 Reproducibility

Toolchain `leanprover/lean4:v4.28.0`, Mathlib commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. Canon repository commit `80d32904d751b45f415e23f872510cc2eddca3ff`. The canon file hashes are listed in the proposals' rationale files. Each proposal (in the outputs of run 2026-10-30) compiles with `lean proposals/P2.`*Name*`.lean`. The sealed Carbon Continuity statement has sha256 `a5a6e8c6...ffded`. To check its proof, run `viridis-lean-check proofs/P2.CarbonContinuity.lean P2.CarbonContinuity`. The sealed Restoration statement has sha256

`fd40b113c06beaed7940b2bf84d931a8b9179dcabaf3c2d27f7300dc27ccff18`, and the canon file it restates has sha256 `667a826a2e5befac67964b69670435362de2d3d792e163807c3bb7c2aec28132`. To check the proof, run `viridis-lean-check proofs/P2.Restoration.lean P2.Restoration`. The sealed Afforestation statement has sha256 `2fc1cbb05d7780f266f9da9c00941ca1b7d8ebc703a11a25fc9bb23237a59436`, and the canon file it restates has sha256 `3ebe33bfd6bd74124e87b5d7e73448db58e1459207fd6ff5c00a90fc3ac9526f`. To check the proof, run `viridis-lean-check proofs/P2.Afforestation.lean P2.Afforestation`. No automated prover jobs (Aristotle) and no external data were used in any of the four sessions.

12 Changelog

- 2026-10-29 (P2-T001): First version. Ported the canon threshold proof to the sealed statement `P2.CarbonContinuity`, and wrote the scope statement.
- 2026-10-30 (P2-T002): Proposed sealed statements `P2.Restoration`, `P2.Afforestation` and `P2.Harmonization`, with rationales, sanity examples and negative controls. Added a method section for writing statements. Condensed the Carbon Continuity material into its own section, and retitled the paper to cover the P2 program.
- 2026-10-31 (P2-T003): Gave a Lean proof of the sealed statement `P2.Restoration`, as a restructured port of the canon proof. Added the Restoration section, covering the proof, its relation to the canon, an unused hypothesis, and the r-versus-n remark. Added a scope paragraph on what the theorem does not settle for the Restoration kernel. Added a status table, and updated the title and abstract.
- 2026-11-01 (P2-T004): Gave a Lean proof of the sealed statement `P2.Afforestation`, porting the canon proof with an explicit squaring step in place of `nlinarith`. Added the Afforestation section (proof, relation to canon, unused hypotheses, the over-promising name), a scope paragraph for the Afforestation kernel, and updated the status table, title and abstract.

References

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